

Probability: Lang. of Uncertainty

- In most of the scientific problems, the quantity of interest remains unknown before it gets observed.
- Probability provides a principled mathematical framework to represent uncertainty.
- Birthday Paradox

$${}^{23}C_2 = 253 \text{ pairs}$$

- Equiprobability Bias - When we know very little, our minds assume that everything is equally likely.
- Probability is not about guessing. It is a mathematical framework based on explicit assumptions.
- Random Experiment - It is a process whose outcome can't be predicted before it is experimented, even though the set of all possible outcomes is known.
- Sample space $\rightarrow (\Omega)$ is a set of all possible outcomes of a random experiment.
- The sample space tells what all can happen, where probability tells the likelihood of each possibility.
- Event - An event is a subset of a sample space. $A \subseteq \Omega$. Probability is assigned to events and not outcomes. $P(A)$ refers to how likely event A can occur.
- Random Variable - It is a funcⁿ $X: \Omega \rightarrow \mathbb{R}$ that assigns a real no. to every outcome in the sample space. It is discrete if we have finite no. of values. It is continuous if it can assume

any value in the interval.

- Probability distribution of a random variable describes its uncertainty.

Discrete -

$$\boxed{p(x) = p(X=x)} \quad \text{s.t. } p(x) \geq 0, \sum_{x \in D} p(x) = 1$$

Continuous -

$$\boxed{f(x) \geq 0, \int_{x \in D} f(x) dx = 1} \quad \text{s.t. } P(a \leq X \leq b) = \int_a^b f(x) dx$$

→ it tells us what values are possible & what is the likelihood of each value.

eg.
$$p(x) = \begin{cases} k & x=0 \\ 2k & x=1 \\ 3k & x=2 \\ 0 & \text{o.t.} \end{cases} \quad k = ?$$

$$k + 2k + 3k = 1 \Rightarrow 6k = 1 \Rightarrow k = \frac{1}{6}$$

- eg. 3 fair coins are tossed. X is the no. of heads. Define X & its PMF

$$X = \{0, 1, 2, 3\}$$

$$p(x) = \begin{cases} 1/8 & 0 \\ 3/8 & 1 \\ 3/8 & 2 \\ 1/8 & 3 \end{cases}$$

HHH
HHT
HTH
HTT
THH
TTH
THT
TTT

- We are mostly interested in the form of RV.

eg.
$$\begin{array}{ccccc} X=x & 0 & 1 & 2 \\ P(X=x) & 0.2 & 0.5 & 0.3 \end{array} \quad Y=X^2.$$

$$\begin{array}{c|cccc}
 Y=y & 0 & 1 & 2 \\
 \hline
 P(Y=y) & 0.2 & 0.5 & 0.3
 \end{array}$$

eg.

$$\begin{array}{c|cccc}
 X=x & -1 & 0 & 1 & 2 \\
 \hline
 P(X=x) & 0.2 & 0.3 & 0.2 & 0.3
 \end{array}$$

$$Y = X^2$$

$$p(y) = \begin{cases} 0 & 0.3 \\ 1 & 0.4 \\ 4 & 0.3 \end{cases}$$

- Bernoulli distribution $\rightarrow$ it models a single experiment with 2 possible outcomes - it can be either success or failure

$$X \sim \text{Bernoulli}(p)$$

$$\boxed{P(X=x) = p^x (1-p)^{1-x}}, \quad x=0,1$$

$$0 < p < 1$$

$$\text{Mean} \Rightarrow E[X] = p$$

$$\text{Variance} \Rightarrow \text{Var}(X) = p(1-p)$$

- Binomial Distribution $\rightarrow$ it models the no. of successes in n independent bernoulli trials

$$X \sim \text{Binomial}(n, p)$$

$$\boxed{P(X=x) = {}^n C_x p^x (1-p)^{n-x}}$$

$$x = 0, 1, \dots, n$$

$$n, 0 < p < 1$$

$$\text{Mean} \Rightarrow E[X] = np, \quad \text{Var} \Rightarrow \text{Var}(X) = np(1-p)$$

- Poisson distribution - no. of events that occurs during a fixed interval of time.

$$X \sim \text{Poisson}(\lambda)$$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$x = 0, 1, \dots$$

$$\lambda > 0; \quad E[X] = \lambda, \quad \text{Var}(X) = \lambda$$

→ it counts independent random event that occurs at a constant avg. time.

- Normal / Gaussian distribution - continuous probability model.

$$X \sim N(\mu, \sigma^2)$$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)}$$

$E[X] = \mu$, $\text{Var}(X) = \sigma^2$, bell-shaped, symmetric around μ , mean = median = mode, total area under the curve = 1.

- $\mu \uparrow \Rightarrow$ distribution shifts towards right, no. shape Δ
- $\sigma^2 \uparrow \Rightarrow$ spread increases, wider curve.

Helpful for normal

- $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$P(Z < -a) = P(Z > a)$$

$$P(Z < a) = 1 - P(Z > a)$$

$$68\% (\mu \pm \sigma), \quad 95\% (\mu \pm 2\sigma), \quad 99\% (\mu \pm 3\sigma)$$

- Expectation - (Mean) summarizes the center or long-run avg. of the probability distribution.

discrete $E[X] = \sum x p(x)$

conti. $E[X] = \int_{-\infty}^{\infty} x f(x) dx$

$$Y = g(X)$$

$$E[Y] = E[g(X)] = \sum_x g(x) p(x)$$

$$\begin{aligned} E[aX + b] &= aE(X) + b \\ E[X + Y] &= E[X] + E[Y] \end{aligned}$$

- Variance tells us the spread of the distribution over mean.

$$\begin{aligned} \text{Var}(X) &= E[(X - E(X))^2] \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

$$SD(X) = \sqrt{\text{Var}(X)}$$

$$\begin{aligned} \text{Var}(X + c) &= \text{Var}(X) \\ \text{Var}(cX) &= c^2 \text{Var}(X) \end{aligned}$$

- In general, expectation tells you the location of the distribution, variance tells its variability.

- Joint dist. $\Rightarrow P(X=x, Y=y)$

Marginal $\Rightarrow P(X=x) = \sum_y P(X=x, Y=y)$

Conditional $P(Y=y | X=x) = \frac{P(X=x, Y=y)}{P(X=x)}$

- 2 events are defined as independent if $P(A \cap B) = P(A)P(B)$

2 RV are independent if,

$$f(x, y) = f_x(x) f_y(y)$$

- Bernoulli (0/1) $\rightarrow$ either success or failure
- Binomial (n success) $\rightarrow$ clinical trials
- Poisson (count) $\rightarrow$ customer arrivals
- Exponential (waiting time) $\rightarrow$ queues
- Normal (continuous measurement)